

Correction

Exercise 1

In the USA, in 2000 a law was passed with the Right To Carry (RTC) arms in certain states (simulated example). US lawmakers are interested in knowing if this increased or not crimes in cities.

Exercise 1

1. Import the data

```
library(rio)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.1
## v lubridate  1.9.4      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
crime<-import("https://github.com/scpo-quantimethods/quantitative_methods_two/raw/main/exercices/exercice1.R")
```

2. Create a time variable (1 if after 2000, 0 if before 2000)

```
crime$time<-ifelse(crime$year>2000,1,0)
```

3. Run a diff-in-diff analysis with crime_rate as the outcome. Do not forget to cluster your SE as you have nested data. Look only at the year before (1999) and the year just after 2001.

```
library(fixest)
did<-feols(crime_rate~time*treated_state,cluster=~state,data=crime[crime$year==1999|crime$year==2001,])
etable(did)
```

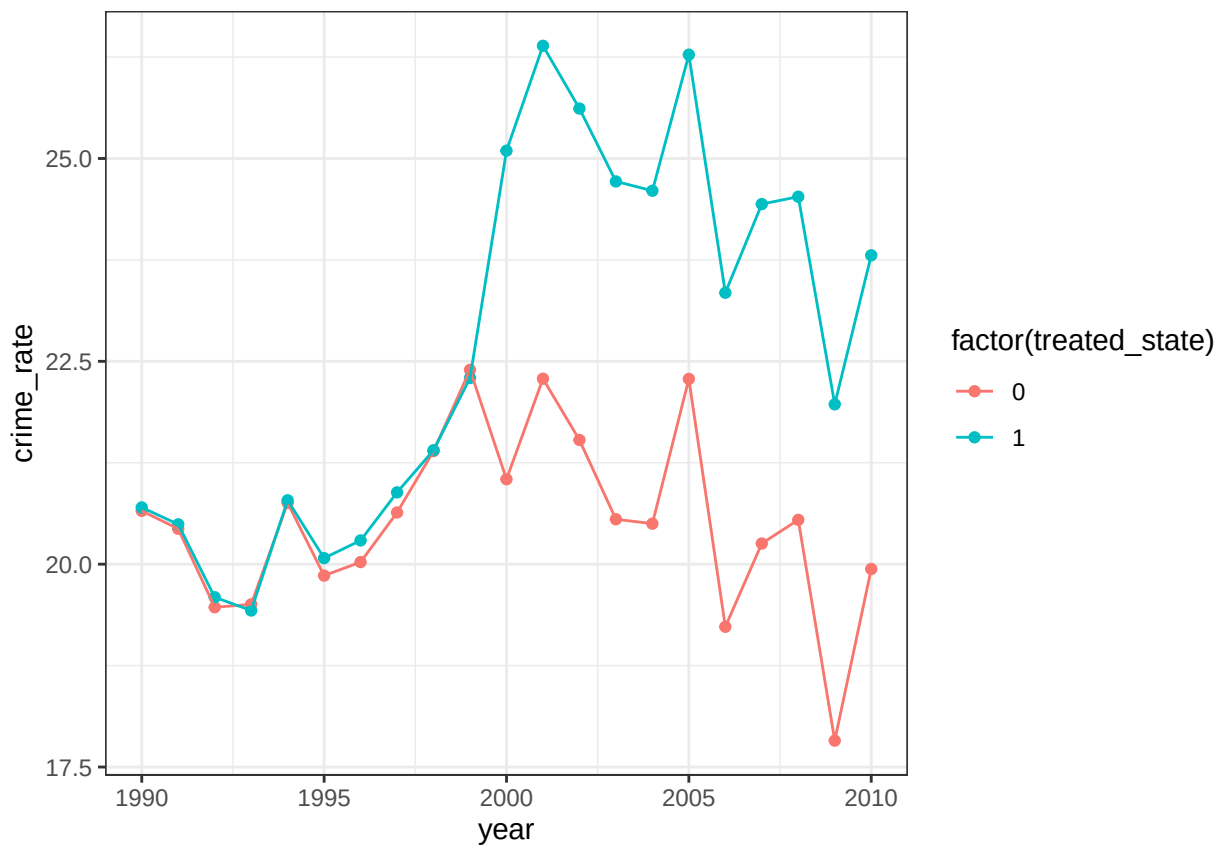
```
##                               did
## Dependent Var.:               crime_rate
##
```

```
## Constant          22.40*** (0.0798)
## time              -0.1098 (0.0714)
## treated_state      -0.1004 (0.1652)
## time x treated_state 4.204*** (0.1114)
## -----
## S.E.: Clustered      by: state
## Observations        5,000
## R2                  0.19950
## Adj. R2             0.19901
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4. What is the effect of RCT policy?

5. Check graphically the parallel trends

```
avg_trends <- aggregate(crime_rate ~ year + treated_state, crime, mean)
library(ggplot2)
ggplot(avg_trends, aes(x=year, y=crime_rate, color=factor(treated_state))) +
  geom_point() +
  geom_line() +
  theme_bw()
```



6. Check with a placebo test the parallel trends (for example between 1997 and 1998)

```
crime$time2<-ifelse(crime$year>1997,1,0)
did2<-feols(crime_rate~time2*treated_state,cluster=~state,data=crime[crime$year==1996|crime$year==1998,])
etable(did2)
```

```
##                                did2
## Dependent Var.:                crime_rate
##
## Constant                      20.03*** (0.0757)
## time2                        1.368*** (0.0609)
## treated_state                 0.2667. (0.1435)
## time2 x treated_state        -0.2555. (0.1280)
## -----
## S.E.: Clustered                by: state
## Observations                  5,000
## R2                           0.04734
## Adj. R2                      0.04677
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

7. Run a diff-in-diff analysis with crime_rate as the outcome. Add population and median income as control variables. Look only at the year before (1999) and the year just after 2001.

```
did3<-feols(crime_rate~time*treated_state+population+median_income,cluster=~state,data=crime[crime$year==1999|crime$year==2001,])
```

```
## Warning: The VCOV matrix is not positive semi-definite and was 'fixed' (see
## ?vcov).
```

```
etable(did3)
```

```
##                                did3
## Dependent Var.:                crime_rate
##
## Constant                      20.52*** (0.3038)
## time                         -0.1292. (0.0693)
## treated_state                 -0.1090 (0.1645)
## population                    1.03e-5*** (1.14e-6)
## median_income                 7.56e-6 (5.86e-6)
## time x treated_state          4.232*** (0.1163)
## -----
## S.E.: Clustered                by: state
## Observations                  5,000
## R2                           0.22524
## Adj. R2                      0.22446
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

8. Compare the results from point 3 and 7.

```
etable(did,did3)
```

```
##                                did                                did3
## Dependent Var.:                crime_rate                crime_rate
##
## Constant                22.40*** (0.0798)    20.52*** (0.3038)
## time                    -0.1098 (0.0714)    -0.1292. (0.0693)
## treated_state           -0.1004 (0.1652)    -0.1090 (0.1645)
## time x treated_state    4.204*** (0.1114)    4.232*** (0.1163)
## population              1.03e-5*** (1.14e-6)
## median_income           7.56e-6 (5.86e-6)
## -----
## S.E.: Clustered          by: state          by: state
## Observations            5,000              5,000
## R2                      0.19950            0.22524
## Adj. R2                 0.19901            0.22446
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

9. Now, use the complete dataset to run your diff-in-diff. However, add fixed effects for both year and states.

```
did4<-feols(crime_rate~time*treated_state|year+state,cluster=~state,data=crime)
```

```
## The variables 'time' and 'treated_state' have been removed because of
## collinearity (see $collin.var).
```

```
etable(did4)
```

```
##                                did4
## Dependent Var.:                crime_rate
##
## time x treated_state    3.632*** (0.0384)
## Fixed-Effects:          -----
## year                    Yes
## state                   Yes
## -----
## S.E.: Clustered          by: state
## Observations            52,500
## R2                      0.28333
## Within R2               0.07634
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Exercise 2

In 2021, a few local governments implemented policies with stricter emissions controls, other local governments have not (simulated example). The air quality data is available from 2018 to 2022.

Exercise 2

1. Import the emissions dataset

```
emissions<-import("https://github.com/scpo-quantimethods/quantitative_methods_two/raw/main/exercices/ex
```

```
## Warning: Missing 'trust' will be set to FALSE by default for RData in 2.0.0.
```

2. Compare the means of pre-treatment (2020) and post-treatment (2022) in both the control and treatment groups.

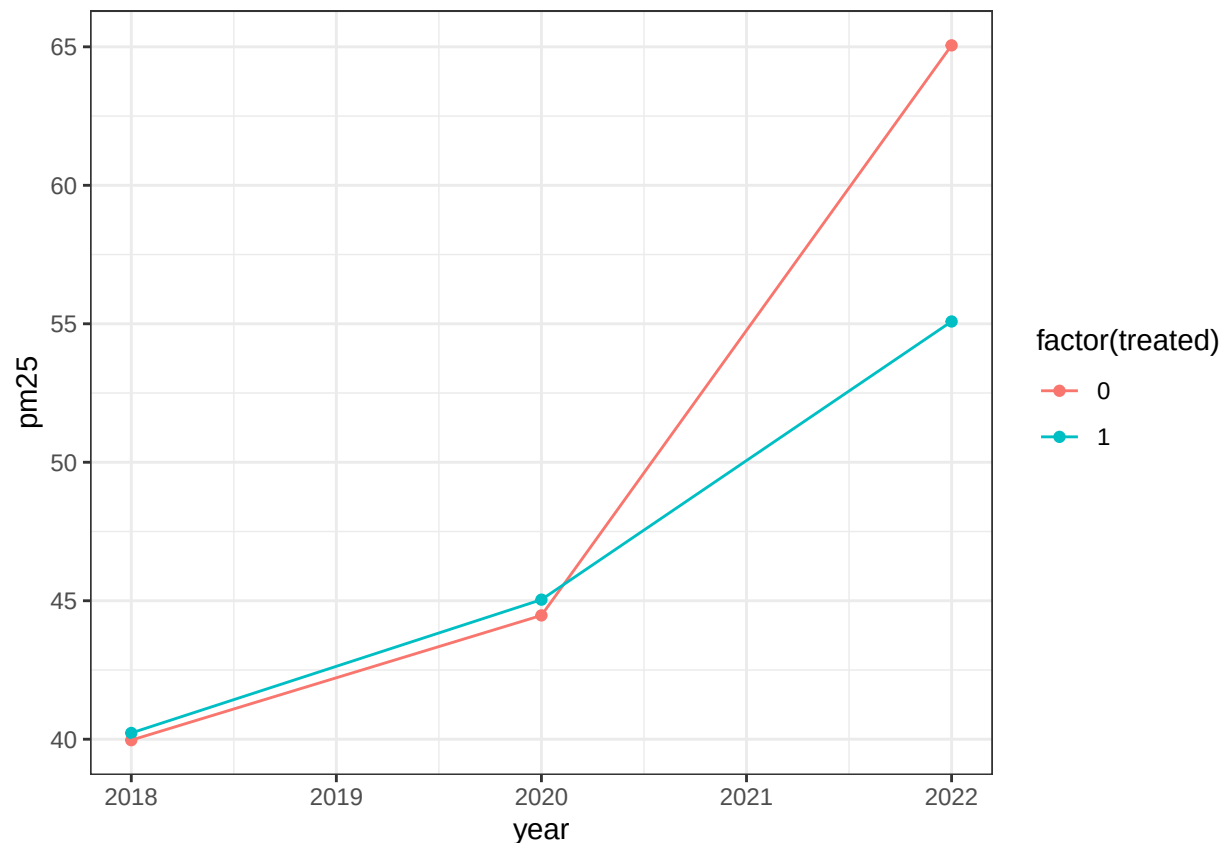
```
emissions%>%  
  filter(year=="2020"|year=="2022")%>%  
  group_by(treated,year)%>%  
  summarize(mean=mean(pm25))
```

```
## 'summarise()' has grouped output by 'treated'. You can override using the  
## '.groups' argument.
```

```
## # A tibble: 4 x 3  
## # Groups:   treated [2]  
##   treated year mean  
##   <dbl> <int> <dbl>  
## 1      0  2020  44.5  
## 2      0  2022  65.1  
## 3      1  2020  45.0  
## 4      1  2022  55.1
```

3. Check out graphically the parallel assumption

```
avg_trends <- aggregate(pm25 ~ year + treated, emissions, mean)  
  
ggplot(avg_trends,aes(x=year,y=pm25,color=factor(treated)))+  
  geom_point()+  
  geom_line()+  
  theme_bw()
```



4. Check out the parallel assumption by running a diff-in-diff between 2018 and 2020

```
did2<-lm(pm25~treated*as.factor(year),data=emissions[emissions$year%in%c("2018","2020"),])
summary(did2)
```

```
##
## Call:
## lm(formula = pm25 ~ treated * as.factor(year), data = emissions[emissions$year %in%
##   c("2018", "2020"), ])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.5123  -3.0751  -0.2226   3.0461  11.0298
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    39.9664     0.7434  53.762 < 2e-16 ***
## treated         0.2595     1.0513   0.247  0.805
## as.factor(year)2020    4.5044     1.0513   4.284 3.19e-05 ***
## treated:as.factor(year)2020  0.3090     1.4868   0.208  0.836
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.702 on 156 degrees of freedom
## Multiple R-squared:  0.2026, Adjusted R-squared:  0.1872
```

```
## F-statistic: 13.21 on 3 and 156 DF, p-value: 9.916e-08
```

5. Produce the difference-in-difference analysis

```
did3<-lm(pm25~treated*post,data=emissions[emissions$year%in%c("2020","2022"),])
summary(did3)
```

```
##
## Call:
## lm(formula = pm25 ~ treated * post, data = emissions[emissions$year %in%
##      c("2020", "2022"), ])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.7370 -3.3261 -0.6279  2.8201 16.1219
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   44.4708     0.7556  58.851  < 2e-16 ***
## treated         0.5685     1.0686   0.532   0.595
## post          20.5821     1.0686  19.260  < 2e-16 ***
## treated:post -10.5380     1.5113  -6.973 8.28e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.779 on 156 degrees of freedom
## Multiple R-squared:  0.7615, Adjusted R-squared:  0.7569
## F-statistic: 166 on 3 and 156 DF, p-value: < 2.2e-16
```

Exercise 3

In 2019, some cities decided to introduce a smoking ban on the terraces of bars and restaurants (simulated example). We want to analyse the effect of the smoking ban on the number of hospitalisations linked to respiratory disease. Use the smoking dataset.

Exercise 3

```
smoking<-import("https://github.com/scpo-quantimethods/quantitative_methods_two/raw/main/exercices/exer
```

```
## Warning: Missing 'trust' will be set to FALSE by default for RData in 2.0.0.
```

1. Compare the means of the outcome variable (called hospital_admissions) between treated and non-treated cities (variable treated) for the years 2018 and 2020 (before and after treatment, variable years or post).

```
smoking%>% group_by(treated,years)%>%
  summarize(mean=mean(hospital_admissions))
```

```
## 'summarise()' has grouped output by 'treated'. You can override using the
## '.groups' argument.
```

```
## # A tibble: 4 x 3
## # Groups:   treated [2]
##   treated years mean
##   <dbl> <chr> <dbl>
## 1      0 2018  99.5
## 2      0 2020  97.3
## 3      1 2018 101.
## 4      1 2020  90.5
```

2. Produce the difference-in-difference analysis

```
did1<-lm(hospital_admissions~treated*post,data=smoking)
summary(did1)
```

```
##
## Call:
## lm(formula = hospital_admissions ~ treated * post, data = smoking)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.0093  -3.2409  -0.4373   2.9408  16.8755
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   99.4922     0.6606 150.606 < 2e-16 ***
## treated        1.0956     0.9437   1.161  0.2471
## post         -2.1625     0.9342  -2.315  0.0217 *
## treated:post  -7.9619     1.3346  -5.966 1.12e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.718 on 196 degrees of freedom
## Multiple R-squared:  0.4112, Adjusted R-squared:  0.4022
## F-statistic: 45.63 on 3 and 196 DF, p-value: < 2.2e-16
```